ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Life Processes in Plants** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** the sunlight leaf	Starch made.
2	**(d)** transpiration	Water vapour condenses.
3	**(b)** taking food from its host	Parasite.
4	**(a)** remain brown	No chlorophyll.
5	**(a)** guard cells	Guard cells swell.
6	**(d)** dead wood	Saprotroph.
7	**(c)** remains brown	No light.
8	**(a)** water and minerals	Large area.
9	**(c)** trapping insects	Gets nitrogen.
10	**(a)** no starch is formed	No CO_2.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	More light for photosynthesis (1); Greater food production (1).	1+1
12	Less water is lost in sunlight (1); Gas exchange continues (1).	1+1
13	Rhizobium fixes nitrogen (1); Soil fertility improves (1).	1+1
14	No photosynthesis (1); Stored starch is used up (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Plant in a closed jar with KOH (1); Control with water (1); Test leaves for starch: only control positive (1).	1+1+1
16	Guard cells swell and the pore opens (1); They shrink and it closes (1); Light and water matter (1).	1+1+1
17	Hydrilla under a funnel in water (1); Test tube over stem (1); Gas collects; glowing splinter relights (1).	1+1+1
	OR: Water goes up via xylem (1); Food moves via phloem (1); Direction differs (1).	1+1+1
18	More light at noon (1); More photosynthesis (1); Less light at dawn (1).	1+1+1
	OR: Rhizobium fixes nitrogen (1); Soil enriched (1); Saves cost (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Broad flat surface (1); Thin (1); Stomata for gases (1); Veins for transport (1); Chloroplasts with chlorophyll (1).	1+1+1+1+1
	OR: Water the plant (1); Cover branch with a clear bag (1); Seal the stem (1); Place in sunlight (1); Droplets inside show transpiration (1).	1+1+1+1+1
20	Potted plant kept in dark 48 h (1); Cover part of a leaf with black paper (1); Expose to sunlight (1); Test with iodine (1); Only uncovered part blue-black (1).	1+1+1+1+1
	OR: Lichen: alga and fungus (1); Alga food, fungus shelter (1); Rhizobium and legumes (1); Plant gives sugar (1); Bacteria fix nitrogen (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that photosynthesis takes place only at night, or that respiration takes place only at night; photosynthesis needs **light**, respiration goes on **all the time**.
- Confusing xylem and phloem: **xylem** = water and minerals up, **phloem** = food from leaves.
- Writing that oxygen is a raw material of photosynthesis; the raw materials are carbon dioxide and water, and oxygen is the *by-product*.
- Mixing up transpiration (loss of water vapour through stomata) with absorption of water by the root hairs.
- Writing that Cuscuta makes its own food; it is a **parasite** without chlorophyll.
- Forgetting that iodine test shows **starch** (blue-black), not glucose, and that the leaf should first be boiled in alcohol to remove chlorophyll.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Life Processes in Plants** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** cells lose water and become limp	Water loss.
2	**(a)** makes food by photosynthesis	Provides food.
3	**(b)** more bubbles in sunlight	Photosynthesis.
4	**(b)** reduce water loss	Conserve water.
5	**(b)** gives shelter and absorbs water	Protection.
6	**(a)** more light	Faster photosynthesis.
7	**(c)** transpiration pull	Suction.
8	**(c)** Rhizobium and pea plant	Symbiosis.
9	**(a)** oxygen	It is a product.
10	**(d)** no photosynthesis and water is saved	Prevent water loss.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It takes food from the host (1); It does not make food (1).	1+1
12	Roots lack oxygen (1); They cannot respire (1).	1+1
13	Leaf becomes pale (1); Blue-black colour is clearly seen (1).	1+1
14	They respire all the time (1); Photosynthesis stops without light (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Leaf forms pitcher (1); Slippery rim and lid (1); Digestive juices (1).	1+1+1
16	Food store (1); Used in germination (1); Energy for seedling (1).	1+1+1
17	Take oxygen from soil air (1); Release CO2 (1); Waterlogged soil blocks it (1).	1+1+1
	OR: Parasite feeds on living host (1); Saprotroph feeds on dead matter (1); Cuscuta vs mushroom (1).	1+1+1
18	Catch more light (1); Compensate low light (1); Increase photosynthesis (1).	1+1+1
	OR: Less evaporation (1); Plants use it for photosynthesis (1); Avoid fungus at night (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Root hair absorbs (1); Xylem lifts (1); Reaches leaves (1); Evaporates at stomata (1); Transpiration (1).	1+1+1+1+1
	OR: Drip irrigation (1); Mulching (1); Fewer stomata in desert plants (1); Spiny leaves (1); Reduce transpiration (1).	1+1+1+1+1
20	Rhizobium in root nodules fixes nitrogen (1); Nitrogen compounds enter the soil (1); Plants absorb them through roots (1); Animals get nitrogen by eating plants (1); Decay returns nitrogen to the soil (1).	1+1+1+1+1
	OR: Root: carrot (1); Stem: potato (1); Leaf: cabbage (1); Seed: wheat (1); Fruit: mango (1).	1+1+1+1+1

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- Writing that oxygen is a raw material of photosynthesis; the raw materials are carbon dioxide and water, and oxygen is the *by-product*.
- Mixing up transpiration (loss of water vapour through stomata) with absorption of water by the root hairs.
- Writing that Cuscuta makes its own food; it is a **parasite** without chlorophyll.
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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Life Processes in Plants** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** parasitism	Host loses food.
2	**(a)** carbon dioxide + water gives glucose + oxygen	Photosynthesis.
3	**(a)** underground stem	Tuber stores starch.
4	**(b)** Venus flytrap	Insectivorous plants.
5	**(d)** glucose + oxygen gives carbon dioxide + water + energy	Opposite.
6	**(b)** root	Tap root.
7	**(d)** fertilisers and manure	Add nutrients.
8	**(b)** a hot, dry, windy day	Dry air.
9	**(d)** stem	Sugar.
10	**(a)** it makes food but needs nitrogen too	Photosynthesis plus extra nitrogen.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	House Rhizobium (1); Fix nitrogen (1).	1+1
12	Loses water by transpiration (1); Absorption cannot keep up (1).	1+1
13	Break dead matter into minerals (1); Soil becomes fertile (1).	1+1
14	Catch more light (1); Exchange gases (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Plants give O2 (1); Animals give CO2 (1); Both use the gases (1).	1+1+1
16	Save water (1); Remove wastes (1); Little sunshine (1).	1+1+1
17	Floating leaves (1); Stomata on upper surface (1); Air spaces in stems (1).	1+1+1
	OR: Nitrogen for proteins (1); Magnesium for chlorophyll (1); Healthy growth (1).	1+1+1
18	Potato stores starch (1); Carrot in the root (1); Seeds store fat (1).	1+1+1
	OR: Xylem brings water (1); Phloem carries food (1); Supports the leaf (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mango: autotroph (1); Cuscuta: parasite (1); Mushroom: saprotroph (1); Pitcher plant: partly heterotroph (1); Lichen: symbiotic (1).	1+1+1+1+1
	OR: Both use glucose (1); Both give CO2 (1); Plants have no lungs (1); Use stomata and root surfaces (1); Energy for life (1).	1+1+1+1+1
20	Plants base of food chains (1); Oxygen release (1); Carbon dioxide absorbed (1); Deforestation reduces both (1); Climate impact (1).	1+1+1+1+1
	OR: Control light (1); Increase CO2 (1); Control temperature (1); Water supply (1); Better yield (1).	1+1+1+1+1

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